AP BIOLOGY TOPIC 7.1

Introduction to Natural Selection

UNIT 7: NATURAL SELECTION (13–20% EXAM WEIGHT)

EVO-1.C | **EVO-1.D** | **EVOLUTIONARY FITNESS**

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 7.1 (INTRODUCTION TO NATURAL SELECTION)

CED Core Vocabulary

Natural selection — is the primary non-random mechanism of adaptive evolution first articulated by Charles Darwin & Alfred Russel Wallace .

2. Phenotypic Variation — Individuals within a population exhibit diverse traits (arising from random mutations, crossing over, independent assortment, and random fertilization).

3. Heritability of Traits — Trait differences must have a genetic basis (DNA alleles) so they can be passed from parents to offspring.

4. Differential Survival & Reproduction — Because resources are limited, individuals with phenotypes better suited to the environment survive at higher rates and leave more fertile offspring.

Survive better — (evade predators, acquire nutrients, tolerate stress)

Reproduce more — (attract mates, produce higher viable fecundity)

Pass on their advantageous alleles — to the next generation

How Natural Selection Affects Populations — Favorable alleles increase in frequency ($\Delta p > 0$) while less favorable phenotypes/alleles decrease over time.

§2. NATURAL SELECTION AS A MAJOR MECHANISM OF EVOLUTION

Core Mechanism

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Natural Selection as a Major Mechanism of Evolution ➔ **Natural selection** ➔ **populations over generations** ➔ **favorable traits (adaptations)**

- **Natural selection** is the primary non-random mechanism of adaptive evolution first articulated by **Charles Darwin & Alfred Russel Wallace**.
- It changes the genetic composition of **populations over generations** when certain heritable traits increase survival and/or reproductive success in a specific environment.
- Over generations, **favorable traits (adaptations)** become more common in the population, while unfavorable traits become less common.

[COMMON AP EXAM FRQ TRAP] **Individuals do NOT evolve; populations do!**

An individual organism cannot alter its own genotype to adapt during its lifetime — evolution is the change in **allele frequencies in a population** across generations.

§3. WHAT NATURAL SELECTION REQUIRES (DARWIN'S 4 POSTULATES)

Prerequisites

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

What Natural Selection Requires (Darwin's 4 Postulates) ➔ **2. Phenotypic Variation** ➔ **3. Heritability of Traits** ➔ **4. Differential Survival & Reproduction**

- **1. Overproduction & Competition for Limited Resources:** Organisms produce more offspring than the environment can support (**N > K**). Individuals compete for scarce **food, water, shelter, territory, and mates**.
- **2. Phenotypic Variation:** Individuals within a population exhibit diverse traits (arising from *random mutations, crossing over, independent assortment, and random fertilization*).
- **3. Heritability of Traits:** Trait differences must have a genetic basis (**DNA alleles**) so they can be passed from parents to offspring.
- **4. Differential Survival & Reproduction:** Because resources are limited, individuals with phenotypes better suited to the environment survive at higher rates and leave more fertile offspring.

[CORE AXIOM / REMEMBER] **Natural selection begins when there are more individuals than resources.** Competition for limited resources causes differential survival and reproduction!

§6. EVOLUTIONARY FITNESS & ADAPTATION

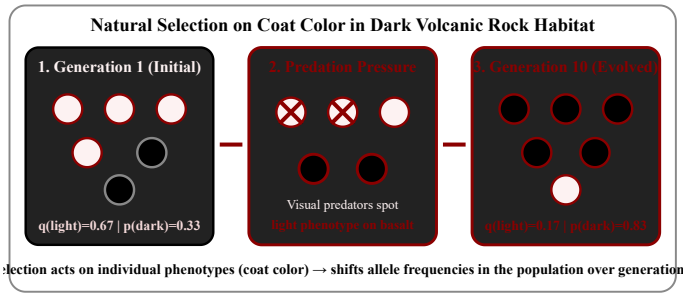
Reproductive Success

- **Evolutionary Fitness = Reproductive Success:** Measured strictly by the **relative number of viable, fertile offspring** an individual contributes to the next generation's gene pool.
  - A "more fit" organism leaves **more surviving offspring** than competitors in the same environment.
  - Fitness is always **relative to the current environment** and context-dependent.
- **Adaptation:** Any **heritable trait** (morphological, physiological, or behavioral) that increases an organism's evolutionary fitness in a particular environment.

[COMMON AP EXAM FRQ TRAP] **Living longer, being stronger, or being bigger does NOT always mean higher fitness!** A sterile mule or a 100-year-old tortoise that leaves 0 offspring has an evolutionary fitness of **zero (w = 0)**.

§7. VISUALIZING NATURAL SELECTION ACROSS GENERATIONS

Phenotype → Allele Shift



§8. ENVIRONMENT DETERMINES WHAT IS FAVORABLE

Biotic & Abiotic Pressures

- A trait is beneficial **only in a specific environmental context**. If the environment shifts, a previously advantageous trait may become neutral or deleterious, and a different trait may be favored.
- **Biotic Factors (Living Selection Pressures):**
  - **Examples:** Predators, interspecific/intraspecific competitors, pathogens/parasites, mate preferences, and mutualists (pollinators).
- **Abiotic Factors (Nonliving Selection Pressures):**
  - **Examples:** Temperature extremes, water availability/drought, sunlight intensity, soil pH/mineral nutrients, salinity, and dissolved oxygen.

[EXAMPLE & BIOLOGICAL APPLICATION] **Rock Pocket Mice (*Chaetodipus intermedius*):** Light sandy fur provides high fitness on light desert granite, but after volcanic eruptions created dark basalt lava flows, the **Mc1r dark fur mutation** became strongly favored on dark rock due to owl predation.

§4. DARWIN'S CORE IDEA: DIFFERENTIAL SURVIVAL & REPRODUCTION

Allele Shifts

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

Darwin's Core Idea ➔ **more favorable phenotypes** ➔ **How Natural Selection Affects Populations**

Heritable Variation → **Resource Competition** → **Differential Reproduction** → **Adaptation Increases**

- Individuals with **more favorable phenotypes** are more likely to:
  - **Survive better** (evade predators, acquire nutrients, tolerate stress)
  - **Reproduce more** (attract mates, produce higher viable fecundity)
  - **Pass on their advantageous alleles** to the next generation
- **How Natural Selection Affects Populations:** Favorable alleles increase in frequency (**$\Delta p > 0$**) while less favorable phenotypes/alleles decrease over time.

[CORE AXIOM / REMEMBER] **Helpful traits stick around because their carriers leave more offspring.**

§5. DARWINIAN SELECTION VS. LAMARCKIAN MISCONCEPTION

FRQ Trap

FEATURE	LAMARCK (INCORRECT MISCONCEPTION)	DARWIN (MODERN EVOLUTIONARY SYNTHESIS)
Mechanism	*Use & Disuse + Inheritance of Acquired Characteristics* during one's lifetime	**Pre-existing heritable variation** + differential reproductive success
Giraffe Neck Example	Ancestral giraffes stretched necks to reach high leaves & passed stretched necks to offspring	Ancestral population already varied in neck length; longer-necked giraffes fed better & left more offspring
Role of Environment	Environment *induces* directed beneficial traits in individuals	Environment **selects** from random, pre-existing genetic variation

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§9. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 7.1 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 7.1: Introduction to Natural Selection ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 1, 2 & 3 (Artificial Selection, Hardy-Weinberg Mathematical Modeling & BLAST Bioinformatics): Comparing allele frequency shifts across generations under directional selection vs. small-population genetic drift simulations, and constructing maximum-parsimony cladograms from BLAST percentage sequence identity matrices (Cytochrome c / β-globin). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Golden Rule: Hardy-Weinberg Equilibrium (HWE): Allele frequencies $p + q = 1$; Genotype frequencies p^2 (homozygous dominant) + $2pq$ (heterozygous carriers) + q^2 (homozygous recessive) = 1. Golden Calculation Sequence: ALWAYS begin with the recessive phenotype frequency (q^2), take $\sqrt{(q^2)}$ to find q, compute $p = 1 - q$, and then calculate $2pq$ (heterozygote carrier frequency). Never take the square root of the dominant phenotype frequency ($p^2 + 2pq$)! **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Evolutionary & Biomedical Disruption Case Study: Incomplete antibiotic regimens or agricultural antibiotic overuse exert strong directional selection on bacterial populations (e.g., MRSA or MDR-Tuberculosis): susceptible cells die while rare, pre-existing resistant mutants (arising via random mutation or R-plasmid conjugation) survive and reproduce. In conservation biology, severe population bottlenecks (e.g., Northern Elephant Seals, African Cheetahs) drastically erode genetic diversity via genetic drift, increasing homozygosity for deleterious recessive alleles and vulnerability to novel pathogens.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 7.1 to prove why the claim holds.

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